Alex Sim

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Narrative

Alex Sim has worked on R&D in data analysis and management fields for scientific and industrial disciplines such as astronomy, climate change simulation, combustion modeling, cosmology, fusion science, genomics, geology, high energy physics, networking, nuclear science, power grid electricity, and behavioral economics, over the last 30 years. His recent research area includes statistical modeling and machine learning methods for autonomic scientific data infrastructure, high-frequency streaming data analysis methods, dynamic resource management, and I/O optimization issues for exascale HPC applications. He has led the system designs, development of algorithms and models, implementation of the systems, management of projects, teams of scientists, postdocs, engineers and/or students, and software package releases under open source license. He has extensive grant proposal writing experience and led projects from the U.S. Department of Energy (DOE) and National Science Foundation (NSF) as a lead Principal Investigator (PI) or Co-PI. He was also involved in technical program committees, steering and advisory committees for conferences, journal editorial boards, and review panels in data, cloud computing, HPC, and networking areas. He was one of the main authors and editors of the standard specification from Open Grid Forum, and contributed to paper publications and technical reports, a number of open source software packages, and multiple patented technologies. He is a senior member of IEEE.

Position History

09/25 - present	Scholarly Engagements Lead, Strategic Engagement Leads, Scientific Data Division, Lawrence Berkeley National Laboratory, Berkeley, CA
07/13 - present	Senior Computing Engineer, Scientific Data Division (former Computational Research Div.), Lawrence Berkeley National Laboratory, Berkeley, CA Job responsibilities: https://sdm.lbl.gov/asim/CSE_Pro_Responsibility_2009.html

12/23 - 09/25	Interim Group Leader, Scientific Data Management Group, Scientific Data Division, Lawrence Berkeley National Laboratory, Berkeley, CA
10/02 - 06/13	Computer Science Engineer IV, Computational Research Division, Lawrence Berkeley National Laboratory, Berkeley, CA
04/00 - 10/02	Computer Science Engineer III, Computational Research Division, Lawrence Berkeley National Laboratory, Berkeley, CA
03/00 - 07/00	Technical Adviser, NetGeo, Inc, Mountain View, CA
07/97 - 04/00	Computer Science Engineer II, National Energy Research Scientific Computing Center, Lawrence Berkeley National Laboratory, Berkeley, CA
10/96 - 11/98	Co-founder, IVision Corporation, Alameda, CA
08/95 - 06/97	Artificial Intelligence Software Engineer, Stottler Henke Associates Inc., San Mateo, CA
03/94 - 08/95	Graduate Research Assistant, Imaging Technologies Group, Lawrence Berkeley National Laboratory, Berkeley, CA
08/90 - 05/93	Graduate Student Instructor, University of California at Berkeley, Berkeley, CA

Education

- M.S. in Computer Science, San Francisco State University, August 1995
- M.A. in Statistics, University of California at Berkeley, May 1993
- B.A. (High Distinction in General Scholarship, equiv. to Magna Cum Laude) in Applied Mathematics and Statistics (Double Major), University of California at Berkeley, May 1990

Awards, Honors and Patents

Best Paper Awards

- Best paper award, "*Time-series Forecasting for Network Utilization in Large-Scale Scientific Workflows*", IEEE International Conference on Computing, Networking and Communications (ICNC 2026), 2026.
- Best paper award, "*Extracting Baseline Electricity Usage Using Gradient Tree Boosting*", International Conference on Big Data Intelligence and Computing (DataCom 2015), 2015.
- Best paper award, "*Grid Collector: Facilitating Efficient Selective Access from Data Grids*", International Supercomputer Conference, Heidelberg, Germany (ISC 2005), 2005.

Patents

- US Patent 10,908,972 B2, "*Methods, systems, and devices for accurate signal timing of power component events*", 2/2021.
- US Patent 10,366,078 B2, "*Data reduction methods, systems and devices*", 7/2019.
- US Patent 8,705,342 B2, "*Co-scheduling of network resource provisioning and host-to-host bandwidth reservation on high-performance network and storage systems"*, 4/2014.

Research Competition Awards

- ICME Research Symposium, Research Poster Competition, 2nd place, Stanford University, 5/2025.
- ACM Student Research Competition (SRC): 1st place SC'23, 2nd place SC'22, 1st place TAPIA'19, 1st place SC'17, 2nd place SC'16, 3rd place SC'16, 3rd place SC'10.
- Data Science Discovery Cloud Computing Award, University of California at Berkeley, 12/2023.
- Federal Laboratory Consortium For Technology Transfer, FAR-WEST Region, Outstanding Partnership Award, *Earth System Grid Federation*, 10/2013.
- Hottest Infrastructure Award, A Data Management Infrastructure for Climate Modeling Research – Super Computing 2000 Network Challenge (SC 2000), 2000.

Institutional Awards and Honors

- Director's Award for Exceptional Scientific Achievement, Lawrence Berkeley National Laboratory, 11/2023.
- Outstanding Performance Award – Lawrence Berkeley National Laboratory (1999, 2001).
- SPOTS Recognition Award – Lawrence Berkeley National Laboratory (1998, 2000, 2015, 2016, 2020).
- Departmental Honors in Statistics – University of California at Berkeley (1990).
- High Distinction in General Scholarship – University of California at Berkeley (1990).
- Undergraduate Scholarships – University of California at Berkeley (1988, 1989).

Grants

G1. *Understanding and Predicting Power and Energy Usage in Large-Scale HPC/AI Models*, LBNL LDRD, ETA/CSA Innovation Project, 1/2025-9/2026, Co-PI, $468K

G2. *OASIS: Open-Source AI Software Infrastructure for Science*, DOE ASCR, 12/2023-9/2025, Co-PI, $870K

G3. *AMASE: Architecture and Management for Autonomic Science Ecosystems*, DOE ASCR, Smart High-Performance Computing and Communications Systems Program, 9/2017-12/2021, Co-PI, $4.2M

G4. *Integration of the Earth System Grid Federation (ESGF) with NERSC HPSS module*, LLNL, 11/2017-10/2018, PI, $130K

G5. *IDEALEM, Novel Data Reduction Based on Statistical Similarity,* LBNL LDRD, CS Innovation Project, 10/2015-9/2016, PI, $100K

G6. *Developing ESGF module for NERSC HPSS*, LLNL, 10/2015-3/2016, PI, $63K

G7. *Scientific Data Services (SDS) – Autonomous Data Management on Exascale Infrastructure*, DOE ASCR, 7/2015-6/2018, Co-PI, $2.4M

G8. *Scalable Analysis Methods and In Situ Infrastructure for Extreme Scale Knowledge Discovery*, DOE ASCR, 10/2014-9/2017, Investigator, $4.5M

G9. *International Collaboration Framework for Extreme Scale Experiments (ICEE)*, DOE ASCR, Scientific Collaborations at Extreme-Scale, 10/2012-9/2015, Co-PI, $1.65M

G10. *Attribute-based Unified Data Access Service,* KISTI, Collaborative research, 4/2012-9/2012, Co-PI, $100K

G11. *Partner Development of Earth System Grid Federation*, DOE BER, subcontract from PCMDI/LLNL, 3/2012-2/2014, PI, $200K

G12. *Advanced Performance Modeling,* DOE ASCR, Terabit Networking Research, 10/2011-9/2014, PI, $1.508M

G13. *Policy-driven Large Scale Data Access Framework*, NSF Collaborative Research SDCI Net, Ref OCI-1127039, 9/2011-8/2014, Co-PI, $1.2M

G14. *Exploration of adaptive network transfer for 100Gbps networks*, DOE ASCR, 4/2011-9/2012, PI, $309K

G15. *Climate100: Scaling the Earth System Grid to 100 Gbps Networks*, DOE ASCR ARRA/ANI, 10/2009-3/2011, Principle Investigator (PI), $201K

Journal Articles and Book Chapters

J1. D. Sung, S. Kim, S. Lee, H. Tang, A. Sim, K. Wu, S. Byna, Y. Son, "*Regen: An Object Layout Regenerator on Large-Scale Production HPC Systems*", Future Generation Computer Systems, Vol. 171, 2025. doi:10.1016/j.future.2025.107830.

J2. R. Frehner, K. Wu, A. Sim, J. Kim, K. Stockinger, "*Detecting Anomalies in Time Series Using Kernel Density Approaches*", IEEE Access, Vol. 12, pp. 33420-33439, 2024. doi:10.1109/ACCESS.2024.3371891.

J3. R. Shao, A. Sim, K. Wu, J. Kim, "*Leveraging History to Predict Abnormal Transfers in Distributed Workflows*", Sensors, 23(12), pp. 5485, MDPI, 2023. doi:10.3390/s23125485.

J4. H-C. Yang, L. Jin, A. Lazar, A. Todd-Blick, A. Sim, K. Wu, Q. Chen, C. A. Spurlock, "*Gender Gaps in Mode Usage, Vehicle Ownership, and Spatial Mobility When Entering Parenthood: A Life Course Perspective*", Systems, 11(6), pp. 314, MDPI, 2023. doi:10.3390/systems11060314.

J5. S. Kim, A. Sim, K. Wu, S. Byna, Y. Son, H. Eom, "*Design and Implementation of I/O Performance Prediction Scheme on HPC Systems through Large-scale Log Analysis*", Journal of Big Data, Vol. 10, Article 65, 2023. doi:10.1186/s40537-023-00741-4.

J6. J. Wang, K. Wu, A. Sim, S. Hwangbo, "*Locating Partial Discharges in Power Transformers with Convolutional Iterative Filtering*", Sensors, 23(4), 1789, 2023, 23, doi: 10.3390/s23041789.

J7. J. Bang, H. Kim, A. Sim, G. Lockwood, H. Eom, H. Sung, "*Design and Implementation of Burst Buffer Over-Subscription Scheme for HPC Storage Systems*", IEEE Access, 11:3386-3401, 2023. doi:10.1109/ACCESS.2022.3233829.

J8. S. Kim, A. Sim, K. Wu, S. Byna, Y. Son, "*Design and Implementation of Dynamic I/O Control Scheme for Large Scale Distributed File Systems*", Cluster Computing, Springer, 2022. doi:10.1007/s10586-022-03640-0.

J9. B. Weinger, J. Kim, A. Sim, M. Nakashima, N. Moustafa, K. Wu, "*Enhancing IoT Anomaly Detection Performance for Federated Learning*", Digital Communications and Networks, Special Issue on Edge Computation and Intelligence, ISSN 2352-8648, 2022. doi:10.1016/j.dcan.2022.02.007. https://www.sciencedirect.com/science/article/pii/S2352864822000190

J10. L. Jin, A. Lazar, C. Brown, V. Garikapati, B. Sun, S. Ravulaparthy, Q. Chen, A. Sim, K. Wu, T. Wenzel, T. Ho, C. A. Spurlock, "*What Makes You Hold on to That Old Car? Joint Insights from Machine Learning and Multinomial Logit on Vehicle-level Transaction Decisions*", Frontiers in Future Transportation, 3:894654, 2022. doi:10.3389/ffutr.2022.894654. https://www.frontiersin.org/articles/10.3389/ffutr.2022.894654/full.

J11. M. Nakashima, A. Sim, Y. Kim, J. Kim, J. Kim, "*Automated Feature Selection for Anomaly Detection in Network Traffic Data*", ACM Transactions on Management Information Systems (TMIS), Vol. 12, No. 3, pp 1-28, 2021. doi:10.1145/3446636.

J12. A. Syal, A. Lazar, J. Kim, A. Sim, K. Wu, "*Network traffic performance analysis from passive measurements using gradient boosting machine learning*", International Journal of Big Data Intelligence, Vol. 8, No. 1, pp 13-30, ISSN: 2053-1389, eISSN: 2053-1397, 2021. doi:10.1504/IJBDI.2021.118741.

J13. L. Jin, A. Lazar, J. Sears, A. Todd, A. Sim, K. Wu, H-C. Yang, C. A. Spurlock, "*Clustering Life Course to Understand the Heterogeneous Effects of Life Events, Gender and Generation on Habitual Travel Modes*", IEEE Access, pp. 1-17, ISSN 2169-3536, 2020. doi:10.1109/ACCESS.2020.3032328.

J14. I. Monga, C. Guok, J. MacAuley, A. Sim, H. Newman, J. Balcas, P. DeMar, L. Winkler, T. Lehman, X. Yang, "*SDN for End-to-end Networked Science at the Exascale*", Future Generation Computer Systems, Vol. 110, pp. 181-201, ISSN 0167-739X, Elsevier, 2020. doi:10.1016/j.future.2020.04.018.

J15. J. Kim, A. Sim, B. Tierney, S. Suh, I. Kim, "*Multivariate Network Traffic Analysis using Clustered Patterns*", Journal of Computing, Vol. 101, No. 4, pp. 339-361, Springer, 2019. doi:10.1007/s00607-018-0619-4.

J16. J. Kim, A. Sim, "*A New Approach to Multivariate Network Traffic Analysis*", Journal of Computer Science and Technology (JCST), Vol. 34, No. 2, pp. 388-402, 2019. doi:10.1007/s11390-019-1915-y.

J17. A. Lazar, L. Jin, C. A. Spurlock, K. Wu, A. Sim, A. Todd. "*Evaluating the Effects of Missing Values and Mixed Data Types on Social Sequence Clustering Using t-SNE Visualization*", ACM Journal of Data and Information Quality, Vol. 11, No. 2, pp. 1-22, 2019. doi:10.1145/3301294.

J18. H. Zhan, G. Gomes, X. S. Li, K. Madduri, A. Sim, K. Wu, "*Consensus Ensemble System for Traffic Flow Prediction*", IEEE Transactions on Intelligent Transportation Systems, Vol. 19, Issue 12, pp. 3903-3914, 2018. doi:10.1109/TITS.2018.2791505.

J19. T. Kim, J. Choi, D. Lee, A. Sim, C. A. Spurlock, A. Todd, K. Wu, "*Predicting Baseline for Analysis of Electricity Pricing*", International Journal of Big Data Intelligence. Special issue on Data to Decision. Vol.5, No.1/2, pp.3-20, 2018. doi:10.1504/IJBDI.2018.10008133.

J20. L. Wu, K. Wu, A. Sim, M. Churchill, J. Choi, A. Stathopoulos, C-S. Chang, S. Klasky, "*Towards Real-Time Detection and Tracking of Spatio-Temporal Features: Blob-Filaments in Fusion Plasma*", IEEE Transactions on Big Data (TBD), Vol. 2, Issue 3, pp. 262-275, Sep. 2016. doi:10.1109/TBDATA.2016.2599929.

J21. W. Yoo, A. Sim, "*Time-series Forecast Modeling on High-Bandwidth Wide Area Network Measurements*", Journal of Grid Computing, Vol. 14, Issue 3, pp 463–476, September 2016, doi:10.1007/s10723-016-9368-9. http://link.springer.com/article/10.1007/s10723-016-9368-9

J22. W. Yoo, M. Koo, Y. Cao, A. Sim, P. Nugent, K. Wu, "*Performance Analysis Tool for HPC and Big Data Applications on Scientific Clusters*", in Conquering Big Data Using High Performance Computing, edited by R. Arora, pp.139-161, Springer International Publishing, 2016, ISBN 978-3-319-33742-5 (ebook), 978-3-319-33740-1 (hardcover). doi:10.1007/978-3-319-33742-5. http://www.springer.com/us/book/9783319337401.

J23. D. H. Bailey, S. Ger, M. L. De Prado, A. Sim, K. Wu, "*Statistical Overfitting and Backtest Performance*", book chapter, "Risk-Based and Factor Investing", edited by Emmanuel Jurczenko, ISTE Press Ltd, Elsevier Ltd, UK, pp. 449-461, ISBN 978-1-78548-008-9, 2015. doi:10.1016/B978-1-78548-008-9.50020-4.

J24. K. Hu, J. Choi, A. Sim, J. Jiang, "*Best Predictive Generalized Linear Mixed Model with Predictive Lasso for High-Speed Network Data Analysis*", International Journal of Statistics and Probability, Vol. 4, No. 2, p132-148, ISSN 1927-7032, E-ISSN 1927-7040, Published by Canadian Center of Science and Education, 2015. doi:10.5539/ijsp.v4n2p132. http://www.ccsenet.org/journal/index.php/ijsp/article/view/48070

J25. D.N. Williams, R. Ananthakrishnan, G. Aloisio, D. Bader, G. Bell, L. Cinquini, S. Denvil, C. Doutriaux, R. Drach, S. Fiore, I. Foster, E. Gonzalez, J. Harney, E. Marques, N. Miller, R. Schweitzer, G. Shipman, C. Silva, A. Sim, K. Taylor, "*Climate Science Responds to "Big Data" Challenges: Accessing and Analyzing Model Output and Observations*", 2012-13 Program on Statistical and Computational Methodology for Massive Datasets, and Bulletin of the American Meteorological Society, 2013

J26. D. N. Williams, A. Sim, et al., "*Earth System Grid Federation: Infrastructure to Support Climate Science Analysis as an International Collaboration. A Data-Driven Activity for Extreme-Scale Climate Science*", Data Intensive Science Series, pp. 121-149, Chapman & Hall/CRC Computational Science, ISBN 9781439881392, 2012. doi:10.1201/b14935-6.

J27. J. Gu, D. Katramatos, X. Liu, V. Natarajan, A. Shoshani, A. Sim, D. Yu, S. Bradley and S. McKee, "*StorNet: Integrated Dynamic Storage and Network Resource Provisioning and Management for Automated Data Transfers*", Journal of Physics: Conf. Ser. vol. 331, 2011. doi:10.1088/1742-6596/331/1/012002.

J28. G Garzoglio, J Bester, K Chadwick, D Dykstra, D Groep, J Gu, T Hesselroth, O Koeroo, T Levshina, S Martin, M Salle, N Sharma, A Sim, S Timm and A Verstegen, "*Adoption of a SAML-XACML Profile for Authorization Interoperability across Grid Middleware in OSG and EGEE*", Journal of Physics: Conf. Ser. vol. 331, 2011. doi:10.1088/1742-6596/331/6/062011.

J29. J. Jensen, R. Downing, D. Ross, A. Sim, "*Practical Grid Storage Interoperation*", Journal of Grid Computing, Vol. 7, Issue 3, pp. 309-317, 2009. doi:10.1007/s10723-009-9127-2.

J30. M. Riedel, E. Laure, T. Soddemann, A. Sim, V. Natarajan, A. Shoshani, J. Gu, et al., "*Interoperation of world-wide production e-Science infrastructures*", Concurrency and Computation: Practice and Experience, Vol. 21, Issue 8, pp. 961-990, 2009. doi:10.1002/cpe.1402.

J31. A. Shoshani, F. Donno, J. Gu, J. Hick, M. Litmaath, A. Sim, "*Dynamic Storage Management*", Book Chapter in Scientific Data Management: Challenges, Technology, and Deployment, Arie Shoshani and

Doron Rotem, Editors, Chapman & Hall/CRC Computational Science, ISBN 9780429189272, 2009. doi: 10.1201/9781420069815.

J32. D. N. Williams, R. Ananthakrishnan, D. E. Bernholdt, S. Bharathi, D. Brown, M. Chen, A. L. Chervenak, L. Cinquini, R. Drach, I. T. Foster, P. Fox, D. Fraser, J. Garcia, S. Hankin, P. Jones, D. E. Middleton, J. Schwidder, R. Schweitzer, R. Schuler, A. Shoshani, F. Siebenlist, A. Sim, W. G. Strand, M. Su, N. Wilhelmi, *"The Earth System Grid: Enabling Access to Multimodel Climate Simulation Data",* American Meteorological Society, Vol. 90, No. 2, 195-205, 2009. doi:10.1175/2008BAMS2459.1.

J33. P. Jakl, J. Lauret, A. Hanushevsky, A. Shoshani, A. Sim and J. Gu, "*Grid data access on widely distributed worker nodes using scalla and SRM*", Journal of Physics: Conference Series, Vol. 119 072019, Distributed Data Analysis and Information Management, 2008. doi:10.1088/1742-6596/119/7/072019.

J34. C. S. Chang, S. Klasky, J. Cummings, R. Samtaney, A. Shoshani, A. Sim, et al., "*Toward a first-principles integrated simulation of tokamak edge plasmas*", Journal of Physics: Conference Series, Vol. 125 012042, 2008. doi:10.1088/1742-6596/125/1/012042.

J35. R Ananthakrishnan, D E Bernholdt, S Bharathi, D Brown, M Chen, A L Chervenak, L Cinquini, R Drach, I T Foster, P Fox, D Fraser, K Halliday, S Hankin, P Jones, C Kesselman, D E Middleton, J Schwidder, R Schweitzer, R Schuler, A Shoshani, F Siebenlist, A Sim, W G Strand, N Wilhelmi, M Su, D N Williams, "*Building a global federation system for climate change research: the earth system grid center for enabling technologies (ESG-CET)*",J ournal of Physics: Conference Series, Vol. 78 012050, 2007. doi:10.1088/1742-6596/78/1/012050.

J36. A. Shoshani, A. Sim, K. Stockinger, *"RRS: Replica Registration Service for Data Grids",* Lecture Notes in Computer Science, Edited by Jean-Marc Pierson, Springer-Verlag GmbH Publisher, Volume 3836, pp. 100-112, 2006. doi:10.1007/11611950_9.

J37. D. E. Middleton, D. E. Bernholdt, D. Brown, M. Chen, A. L. Chervenak, L. Cinquini, R. Drach, P. Fox, P. Jones, C. Kesselman, I. T. Foster, V. Nefedova, A. Shoshani, A. Sim, W. G. Strand and D. Williams, "Enabling worldwide access to climate simulation data: the earth system grid (ESG)", Journal of Physics: Conference Series Vol. 46 510, 2006. doi:10.1088/1742-6596/46/1/070.

J38. D. Bernholdt, S. Bharathi, D. Brown, K. Chanchio, M. Chen, A. Chervenak, L. Cinquini, B. Zrach, I. Foster, P. Fox, J. Garcia, C. Kesselman, R. Markel, D. Middleton, V. Nefedova, L. Pouchard, A. Shoshani, A. Sim, G. Strand, D. Williams, "*The Earth System Grid: Supporting the Next Generation of Climate Modeling Research*", IEEE Vol. 93, No. 3, pp. 485-495, 2005. doi:10.1109/JPROC.2004.842745.

J39. A. Shoshani, A. Sim, J. Gu, "*Storage Resource Managers: Essential Components for the Grid*", Grid Resource Management: State of the Art and Future Trends, Edited by Jarek Nabrzyski, Jennifer M. Schopf, Jan Weglarz, Kluwer Academic Publishers, pp. 321–340, ISBN 1402075758, 2003. doi:10.5555/976113.976134.

J40. A. L. Chervenak, E. Deelman, C. Kesselman, W. E. Allcock, I. T. Foster, V. Nefedova, J. Lee, A. Sim, A. Shoshani, B. Drach, D. Williams, D. Middleton, "*High-performance remote access to climate simulation data: a challenge problem for data grid technologies*", Journal of Parallel Computing, Vol. 29, Issue 10, pp. 1335-1356, 2003. doi:10.1016/j.parco.2003.06.001.

J41. L. Bernardo, B. Gibbard, D. Malon, H. Nordberg, D. Olson, R. Porter, A. Shoshani, A. Sim, A. Vaniachine, T. Wenaus, K. Wu, D. Zimmerman, "*New Capabilities in the HENP Grand Challenge Storage Access System and its Application at RHIC*", Journal of Computer Physics Communications, Vol. 140, Issues 1–2, pp. 179-188, 2001. doi:10.1016/S0010-4655(01)00269-7.

J42. A. Sim, H. Nordberg, L.M. Bernardo, A. Shoshani and D. Rotem, "*Experience with using CORBA to implement a file caching coordination system*", Concurrency and Computation: Practice and Experience, Vol. 13, issue 7, pp. 605-619, 2001. doi:10.1002/cpe.561.

J43. A. Sim, B. Parvin, P. Keagy, “*Invariant representation and hierarchical network for inspection of nuts from X-ray images*”, International Journal of Imaging Systems and Technology, Vol. 7, Issue 3, pp. 231-237, 1996. doi:10.1002/(SICI)1098-1098(199623)7:3<231::AID-IMA11>3.0.CO;2-1.

Conference Publications - Refereed

C1. C. Kim, A. Sim, K. Wu, H. Tang, Y. Son, J. Park, S. Kim, "*AURORA-Q: Asynchronous Unified Resource Optimizer for Quantum Simulation on Leadership-Scale HPC System*", IEEE International Conference on Distributed Computing Systems (ICDCS 2026), 2026.

C2. C. Kim, E. Sohn, S. Kim, A. Sim, K. Wu, H. Tang, Y. Son, S. Kim, "*ScaleQsim: Highly Scalable Quantum Circuit Simulation Framework for Exascale HPC Systems*", ACM SIGMETRICS, Vol. 9, No. 3, Article 62, 2026. doi:10.1145/3771577.

C3. V. Yakubovich, I. Sim, P. Kim, N. Monozon, J. Chung, A. Sim, K. Wu, "*Time-series Forecasting for Network Utilization in Large-Scale Scientific Workflows*", IEEE International Conference on Computing, Networking and Communications (ICNC 2026), 2026. Best Paper Award. doi:10.1109/ICNC68183.2026.11417004.

C4. M. Sudarshan, A. Sim, K. Wu, "*Predicting Dataset Popularity for Improved Distributed Content Caching in High Energy Physics*", IEEE International Conference on Computing, Networking and Communications (ICNC 2026), 2026. doi:10.1109/ICNC68183.2026.11416999.

C5. I. Mahmud, K. Wu, A. Sim, A. Mandal E. Deelman, "*BBRv3 Startup Behavior: Analysis and Fairness Enhancements*", IEEE International Conference on Computing, Networking and Communications (ICNC 2026), 2026. doi:10.1109/ICNC68183.2026.11416923.

C6. I. Mahmud, K. Wu, A. Sim, A. Mandal E. Deelman, "*Validating TCP Behavior in DISTRI: A Comparison of Simulated and Real-World Network Performance for Distributed Computing*", IEEE International Conference on Computing, Networking and Communications (ICNC 2026), 2026. doi:10.1109/ICNC68183.2026.11416891.

C7. C. M. Oguchi, V. K. Lakshminarayana, A. Sim, K. Wu, D. Ghosal, "*Achieving Deterministic and Reliable Large-Scale Data Transfers in a Scientific Network*", IEEE Military Communications Conference (MILCOM 2025), 2025. doi:10.1109/MILCOM64451.2025.11309880.

C8. A. Sim, E. Wang, R. Monga, K. Wu, C. Guok, I. Monga, D. Davila, F. Wurthwein, J. Balcas, H. Newman, "*Comparing Cache Utilization Trends for Regional Data Caches*", European Physical Journal (EPJ) Web of Conferences (CHEP 2024), Vol. 337, No. 01341, 2025. doi:10.1051/epjconf/202533701341.

C9. J. Aldrich, A. Sim, K. Wu, S. Yoo, H. Ito, V. Garonne, E. Lancon, "*Exploring Data Caching Policy with Data Access Patterns from dCache Logs*", European Physical Journal (EPJ) Web of Conferences (CHEP 2024), Vol. 337, No. 01340, 2025. doi:10.1051/epjconf/202533701340.

C10. S. Kim, S. Kim, C. Kim, A. Sim, K. Wu, H. Tang, "*SWIFTN: Accelerating Quantum Circuit Simulation Through Tensor Optimization*", 25th IEEE International Symposium on Cluster, Cloud, and Internet Computing (CCGrid 2025), 2025. doi:10.1109/CCGRID64434.2025.00022.

C11. J. Kim, A. Sim, K. Wu, J. Kim, "*Improving Slow Transfer Predictions: Generative Methods Compared*", IEEE International Conference on Computing, Networking and Communications (ICNC 2025), 2025. doi:10.1109/ICNC64010.2025.10994006.

C12. B. Fan, A. Sim, K. Wu, J. Kim, "*Conditional Recurrent Neural Networks for Enhancing Throughput Prediction and Slow File Transfers Detection in Large Science Workflows*", 22nd IEEE Consumer Communications & Networking Conference (CCNC 2025), 2025. doi:10.1109/CCNC54725.2025.10976138.

C13. B. Dong, A. Nayak, K. Wu, V. Tribaldos, J. Ajo-Franklin, Q. Zhang, S. Byna, F. Guo, P. Dobson, A. Sim, "*TensorSearch: Parallel Similarity Search on Tensors*", IEEE International Conference on Big Data (BigData), 2024. doi:10.1109/BigData62323.2024.10825798

C14. M. Schreyer, T. Sattarov, A. Sim, K. Wu, "*Imb-FinDiff: Conditional Diffusion Models for Class Imbalance Synthesis of Financial Tabular Data*", 5th ACM International Conference on AI in Finance (ICAIF2024), 2024. doi:10.1145/3677052.3698659

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D4. *SENSE: Intelligent Network Services for Science Workflows*, International Conference for High Performance Computing, Networking, Storage and Analysis (SC'19), 2019.
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D13. *Berkeley Storage Manager (BeStMan)*, Super Computing (Tampa, FL), 2006
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D25. *Robust File Replication of Massive Datasets on the Grid*, Super Computing (Baltimore, MD), 2002
D26. *GridFTP-HPSS Access Provided through HRM*, Super Computing (Baltimore, MD), 2002
D27. *Scientific Data Management: Middleware support for High-Level Requests to Distributed Data*, Super Computing (Denver, CO), 2001
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Projects

Lawrence Berkeley National Laboratory

1. CS ESnet Program – Data Strategy 1/2019 - present
   DOE Office of Science
2. Understanding and Predicting Power and Energy Usage in Large-Scale HPC/AI Models
   LBNL LDRD ETA/CSA Innovation Project 1/2025-9/2026
3. OASIS: Open-Source AI Software Infrastructure for Science 12/2023-9/2025
   DOE ASCR
4. RAPIDS2: a SciDAC Institute for Resource and Application Productivity through computation, Information, and Data Science 9/2020 - 9/2025
   DOE ASCR
5. ExaFEL: Data Analytics at the Exascale for Free Electron Lasers. 9/2020 – 9/2021
   DOE ASCR
6. ExaIO: Delivering Efficient Parallel I/O on Exascale Computing Systems with HDF5 and Unify (ECP Software Technology project) 10/2019 - 9/2023
   DOE ASCR
7. ExaLearn: Codesign Center for Exascale Machine Learning Technologies 9/2018 - 8/2022
   DOE ASCR
8. Scientific Data Management Research in the Exascale Era 10/2018 - 9/2021
   DOE ASCR
9. ExaFEL: Data Analytics at the Exascale for Free Electron Lasers 10/2017 - 9/2020
   DOE ASCR
10. SENSE: SDN for End-to-end Networked Science at the Exascale 10/2017 - 3/2019
    DOE ASCR
11. RAPIDS: a SciDAC Institute for Resource and Application Productivity through computation, Information, and Data Science 9/2017 - 9/2020
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12. Integration of the ESGF with NERSC HPSS module 10/2017 - 9/2018
    Lawrence Livermore National Laboratory
13. AMASE: Architecture and Management for Autonomic Science Ecosystems 9/2017 - 8/2021
    DOE ASCR
14. Exascale Computing Project Chombo IO study and optimization 1/2017 - 12/2019
    DOE ASCR
15. Exascale Computing Project AMReX IO study and optimization 1/2017 - 12/2019
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18. IDEALEM: Novel Data Reduction Based on Statistical Similarity 10/2015 - 9/2017
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19. Developing ESGF module for NERSC HPSS 10/2015 - 3/2016
    Lawrence Livermore National Laboratory
20. Open Framework for High-Performance Streaming Analytics 10/2015 - 9/2018
    LBNL LDRD
21. Behavior Analytics 6/2015 - 8/2016
    LBNL LDRD
22. Scalable Analysis Methods and In Situ Infrastructure for Extreme Scale Knowledge Discovery
    DOE ASCR 10/2014 - 9/2017
23. Scientific Data Management Research in the Exascale Era 12/2014 - 6/2018
    DOE ASCR base research
24. X-SWAP: Extreme-Scale Scientific Workflow Analysis and Prediction 10/2014-9/2017
    DOE ASCR Next Generation Networking Program
25. SciDAC Scalable Data management, Analysis and Visualization Institute 3/2012 - 2/2017
    DOE ASCR SciDAC
26. International Collaboration Framework for Extreme Scale Experiments (ICEE) 10/2012 - 9/2015
    DOE ASCR Scientific Collaborations at Extreme-Scale
27. Attribute-based Unified Data Access Service 4/2012 - 9/2012
    KISTI collaborative research project
28. Partner Development of Earth System Grid Federation 3/2012 - 2/2014
    DOE BER PCMDI/LLNL collaboration project
29. APM: Advanced Performance Modeling 10/2011 – 1/2015
    DOE Terabit Networking Research project
30. ADAPT: Adaptive Data Access and Policy-driven Transfers 8/2011 - 1/2015
    NSF SDCI Net project
31. Climate100: Scaling the Earth System Grid to 100 Gbps Networks 9/2009 - 9/2012
    DOE ANI project
32. StorNet: End-To-End Resource Provisioning and Management for Automated Data Transfer
    DOE network research project 8/2009 - 9/2011
33. Storage Resource Manager 4/2007 - 9/2011
    DOE base research program project
34. Open Science Grid: SRM-Tester 9/2006 - 12/2011
    DOE Science Discovery through Advanced Computing II project
35. Scaling the Earth System Grid to Petascale Data 9/2006 - 9/2011
    DOE Science Discovery through Advanced Computing II project
36. Open Science Grid: DRM deployment activity group 6/2004 - 8/2008
37. OASIS: WSRF 6/0205 - 4/2007
38. Storage Resource Management for Data Intensive Grid Applications 7/2001 - 9/2006
    DOE National Collaboratories Middleware Technology project
39. Particle Physics Data Grid 7/2001 - 9/2006
    DOE Science Discovery through Advanced Computing project
40. Earth Systems Grid II 4/2001 - 9/2006
    DOE Science Discovery through Advanced Computing project
41. Scientific Data Management 7/2001 - 9/2006
    DOE Science Discovery through Advanced Computing project
42. Earth Systems Grid, DOE Next Generation Internet project 7/1999 - 3/2001
    DOE BER

43. Particle Physics Data Grid, DOE Next Generation Internet project 6/1999 - 06/2001
    DOE ASCR/HEP
44. High Energy and Nuclear Physics Grand Challenge 10/1997 - 03/2001
    DOE ASCR
45. 3-D geometric modeling and visualization for computational fluid dynamics 7/1997 - 10/1997
    DOE ASCR
46. Machine Vision Inspection and Classification System 3/1994 - 8/1995
    USDA

NetGeo Inc. 3/2000 - 7/2000
1. Aggregation methods for IP-based geographical information
2. Probabilistic validation methods for uncertainly information
3. Web-based infrastructure design for fast retrieval of the geographical information

IVision Corporporation 10/1996 - 11/1998
1. System design for real-time image tracking and capturing device
2. Managerial responsibilities for running the company

Stottler Henke Associates, Inc. 8/1995 - 6/1997
1. Automatic Design of Steel Manufacturing Processes
2. Intelligent Tutoring System (ITS) for Long-Range Acoustic Detection of Submarines
3. Parts Obsolescence Prediction
4. Portable General Naval Tactics Representation System
5. Intelligent Tutoring System for AEGIS Total Ship Survivability Team Training
6. Piloted Approach Decision Aid Logic System
7. Intelligent Training System for a Dismounted Infantry Virtual Reality Environment
8. Environmental Knowledge Base System

Activities

Technical Program/Standard Committee
- AI-Science 2018 - 2019 (ACM Workshop in Autonomous Infrastructure for Science)
- AICT 2016 - 2021 (Advanced International Conference on Telecommunications)
- BDCAT 2017 - 2025 (IEEE/ACM International Conference on Big Data Computing, Applications and Technologies)
- CCGrid 2021 (IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing)
- CCSNA 2014 - 2018 (IEEE Cloud Computing Systems, Networks, and Applications Workshop)
- Cloud Computing 2014 - 2025 (International Conference on Cloud Computing, GRIDs, and Virtualization)
- CloudNet 2021 - 2023 (IEEE International Conference on Cloud Networking)
- Cluster 2018 (IEEE Cluster Conference)
- DADC 2008 - 2009 (Workshop on Data-aware Distributed Computing in conjunction with HPDC)
- Data Analytics 2022-2025 (International Conference on Data Analytics)
- DDINS 2022-2025 (IEEE International Workshop on Data Driven Intelligence for Networks and Systems)
- DIDC 2016 (International Workshop on Data-intensive Distributed Computing)
- DISE1 2018 (Workshop on Deep Learning for Safety-Critical in Engineering Systems)

- FGCN 2008 (International Conference on Future Generation Communication and Networking)
- GDC 2009 (Symposium on Grid and Distributed Computing)
- Globecom 2021 - 2023 (IEEE Global Communications Conference)
- Grid 2008, 2010 - 2012 (IEEE International Conference on Grid Computing)
- HPC Asia 2022 (International Conference on High Performance Computing in Asia-Pacific Region)
- ICC 2023 - 2026 (IEEE International Conference on Communications)
- ICCN 2019 - 2022 (IEEE International Workshop on Intelligent Cloud Computing and Networking)
- ICNS 2016 - 2021, 2023-2025 (International Conference on Networking and Services)
- ICONS 2023 (International Conference on Systems)
- ICT 2026 (IEEE International Conference on Telecommunications)
- ISAV 2015 - 2016 (Workshop on In Situ Infrastructures for Enabling Extreme-Scale Analysis and Visualization)
- ISCC 2017 - 2020 (IEEE Symposium on Computers and Communication)
- NAS 2020 - 2022 (IEEE International Conference on Networking, Architecture, and Storage)
- NDM 2011 (International Workshop on Network-Aware Data Management)
- NMIC 2019 - 2020 (International Workshop on Network Meets Intelligent Computations)
- OASIS WSRF 2004 - 2007 (Web Services Resource Framework Technical Committee)
- OASIS WSN 2004 - 2007 (Web Services Notification Technical Committee)
- OGF GSM 2003 - 2009 (Open Grid Forum Grid Storage Management Working Group)
- SC 2004 (ACM/IEEE The International Conference for High Performance Computing, Networking, Storage, and Analysis)
- SIoTEC 2024 (IEEE Workshop on Secure IoT, Edge and Cloud systems)
- SNTA 2019 - 2024 (ACM Workshop on Systems and Network Telemetry and Analytics)
- SNTA 2018 (IEEE Workshop on Scalable Network Traffic Analytics)
- UCC 2011 - 2025 (IEEE International Conference on Utility and Cloud Computing)
- VIRT 2018 - 2021 (Special Session on Virtualization in Machine Learning, High Performance Computing and Simulation)
- WATCH 2025 (International Workshop on Analytics, Telemetry, and Cybersecurity for HPCC)

Chair, Steering and Advisory Committee

- Chair - International Workshop on Autonomous Infrastructure for Science (AI-Science), ACM, 2018-2019
- Chair - International Conference on High Performance Computing in Asia-Pacific Region (HPC Asia), Architecture and Network Areas, 2022
- Chair - International Workshop on Scalable Network Traffic Analytics (SNTA), IEEE, 2018
- Chair - International Workshop on Systems and Network Telemetry and Analytics (SNTA), ACM, 2019-2022
- Founding member - IEEE SIG-IIE (Special Interest Group on Intelligent Internet Edge), 2019-2025
- Founding member - Open Grid Forum Grid Storage Management Working Group, 2003-2009
- Steering Committee - International Conference on Cloud Computing, GRIDs, and Virtualization (Cloud Computing), IARIA, 2017-2025
- Steering Committee - International Conference on Networking and Services (ICNS), IARIA, 2022-2025
- Steering Committee - International Workshop on Analytics, Telemetry, and Cybersecurity for HPCC (WATCH), ACM, 2025
- Steering Committee - International Workshop on Systems and Network Telemetry and Analytics (SNTA), ACM, 2022-2024

- Industry/Research Advisory Committee - International Conference on Networking and Services (ICNS), IARIA, 2017-2021

Journal editorial board

- Frontiers in High Performance Computing, Frontiers (2023-2025)
- International Journal of Grid and Distributed Computing, Nadia (2013-2025)

Journal guest editor

- International Journal of Big Data Intelligence, Special Issue on Systems and Network Telemetry and Analytics, Inderscience (2019-2022)
- Sensors, Special Issue on Anomaly Detection and Monitoring for Networks and IoT Systems, MDPI (2022-2024)

Journal review

- IEEE Network (2021)
- IEEE Transactions on Parallel and Distributed Systems, in the special issue on Many-Task Computing (2010)
- Journal of Computing, Springer (2020)
- Journal of Machine Learning, Springer (2019)
- Journal of Parallel and Distributed Computing, Elsevier (2014-2015)
- Journal of the Network and Systems Management, Springer (2013)

Grants review

- ASCR Leadership Computing Challenge (ALCC) (2023)
- DOE SBIR-STTR Phase I Grant (2005,2007,2010,2011,2018,2020,2021)
- DOE SBIR-STTR Phase II Grant (2011,2019, 2022)
- Dutch Research Council (NWO), Domain Science (ENW), Open Competition, KLEIN (2020)
- LBNL Laboratory Directed Research and Development (LDRD) (2023, 2025)
- National Science Foundation (NSF) proposal - Office of Cyberinfrastructure (OCI) (2012)

Laboratory internal service

- Student supervisor (undergrads, grads, high school students), LBNL
- Postdoc supervisor, LBNL
- Alvarez Postdoctoral Fellowship Search Committee, LBNL, 2010
- Engineering staff search/interview committee, ESnet, 2022-2023
- Experiences in Research in K-12 education programs, LBNL, 2025
- Job Fair volunteer, LBNL, 2020
- Postdoc search/interview committee, ESnet, 2018-2023
- Postdoc search/interview committee, LBNL, 2012, 2015-2023
- Research Scientist search/interview committee, LBNL, 2014-2019, 2023
- Senior Computing Engineer application review committee, Chair, LBNL, 2022-2025
- Senior Computing Engineer application review committee, LBNL, 2015-2022
- Software Engineering and Management Committee (SEMC), LBNL, 2014-2017

Society membership

- Senior Member of Institute of Electrical and Electronics Engineers, Inc (IEEE)
- Member of Association for Computing Machinery (ACM)
- Member of Society of Actuaries
- Honorary Rosalind Membership of London Journal Press, ID #IF19589, 2020

Miscellaneous

- U.S. Citizen

- U.S. DoD security clearance – secret level (1995-1997)
- Contributions to the Nobel Peace Prize 2007 to the Intergovernmental Panel on Climate Change (IPCC) with climate data management
- Contributions to the Nobel Prize in Physics 2017 to physicists from the Laser Interferometer Gravitational-Wave Observatory (LIGO) collaboration project with the data management software, Berkeley Storage Manager (BeStMan) managing scientific data storage
- Albert Nelson Marquis Lifetime Achievement Award, 2018
- Marquis Who’s Who in the World (2019-2021), Who’s Who in America (1999,2003), Who's Who in the World, Lifetime achievement (2020)
- IARIA certificate of appreciation: for contribution to Information Processing Panel during DataSys 2020